Urban Land-Use Efficiency: A Stable Metric for SDG 11.3.1 and Its Economic Drivers

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Abstract

The official SDG 11.3.1 indicator, the ratio of land consumption rate to population growth rate (LCRPGR), is unstable: it divides by a growth rate that approaches zero and mixes an arithmetic numerator with a logarithmic denominator. We propose the Built-up per Capita Rate (BpCR), the log change in built-up area per person, which equals the difference of two log growth rates and is therefore well-defined for every country-period. Using a harmonised global panel of 193 UN member states (1985–2020) built from GHSL built-up and population grids under the Degree of Urbanisation, we (i) document where LCRPGR breaks down, (ii) show BpCR reproduces the policy signal without the instability, and (iii) estimate its economic drivers with two-way fixed-effects and dynamic-panel GMM models.

Keywords: SDG 11.3.1, Urban land-use efficiency, Built-up per capita rate (BpCR), LCRPGR, Dynamic panel GMM, GHSL / Degree of Urbanisation

1. Introduction

Cities must grow to house more people; the policy question is how much land each new resident consumes (Seto et al., 2011; Oueslati et al., 2015). SDG target 11.3 asks countries to keep land consumption in line with population growth, and indicator 11.3.1 operationalises this as the LCRPGR ratio (UN-Habitat, 2025).

2. The SDG 11.3.1 indicator and its limitations

$$\mathrm{LCRPGR} = \frac{\mathrm{LCR}}{\mathrm{PGR}}, \qquad \mathrm{LCR} = \frac{V_t - V_{t-z}}{V_{t-z}} \cdot \frac{1}{z}, \qquad \mathrm{PGR} = \frac{\ln(P_t/P_{t-z})}{z}.$$

The arithmetic LCR over a logarithmic PGR makes the ratio asymmetric, and PGR in the denominator makes it diverge when population is roughly stable (Nicolau et al., 2019).

3. A stable metric: the Built-up per Capita Rate (BpCR)

$$\mathrm{BpCR}_{i,t} = \frac{1}{z} \ln\left(\frac{V_t/P_t}{V_{t-z}/P_{t-z}}\right) = \mathrm{LCR}_{\log} - \mathrm{PGR}_{\log}.$$

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4. Data

See Pesaresi et al. (2024), Schiavina et al. (2023), UN SNAAMA (2025) and UN DESA WPP (2024) for the source datasets.

Table 1: Estimation sample

Quantity	Value
Countries	193
Country-period observations	2116

5. Empirical strategy

$$\text{Metric}_{i,t} = \rho\,\text{Metric}_{i,t-1} + \beta_1 \ln(\text{UrbDens})_{i,t} + \beta_2\,\text{UrbShare}_{i,t} + \beta_3 \ln(\text{GDPpc})_{i,t} + \beta_4\,\text{NetMigr}_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}.$$

6. Results

7. Robustness

8. Discussion

9. Conclusion

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